



NORTH SYDNEY
GIRLS HIGH SCHOOL

2009

HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK #3

Mathematics Extension 1

Student Name: _____ Teacher: _____

General Instructions

- Reading time – 2 minutes.
- Working time – 65 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for untidy or badly arranged work.
- Start each **NEW** question on a separate answer sheet.

Total Marks: 57 Marks

- Attempt Questions 1 - 6
- All questions are NOT of equal value.

	Q 1		Q 2		Q 3		Q4			Q5	Q6			Total
	acd	b	abcd	e	a	b	a	b	c		a	b	c	
H 6														/3
H 8														/10
H 9														/31
HE 2														/3
HE 6														/10
	/9		/10		/10		/9			/10	/9			/57

Total marks – 57

Attempt Questions 1 - 6

All questions are NOT of equal value

Start each question on a SEPARATE answer sheet.

Question 1	(9 marks)	Marks
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(a)	Evaluate $\int_0^1 \frac{dx}{2x+1}$, leaving your answer in the exact form.	2
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(b)	Using the substitution $u = 4 - x^2$, evaluate $\int \frac{x}{\sqrt{4-x^2}} dx$	3
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(c)	Let $f(x) = \frac{1}{2}(e^x + e^{-x})$ and $F(x) = \frac{1}{2}(e^x - e^{-x})$ Prove that $[f(x) + F(x)]^n = f(nx) + F(nx)$	2
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(d)	Evaluate $\int_0^1 \frac{e^x}{e^x + 1} dx$	2
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Question 2	(10 Marks) Start a NEW answer sheet.	Marks
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(a) Solve $e^x = 5$, leaving your answer correct to 3 decimal places **1**

(b) Find a primitive of $\frac{3x}{1+x^2}$ **2**

(c) Find $\frac{d}{dx}(3x \log_e x)$ **2**

(d) Evaluate $\int_0^3 3^x dx$ **2**

(e) Using the substitution $u = \log_e x$, evaluate $\int_1^e \frac{(1 + \log_e x)^2}{x} dx$ **3**

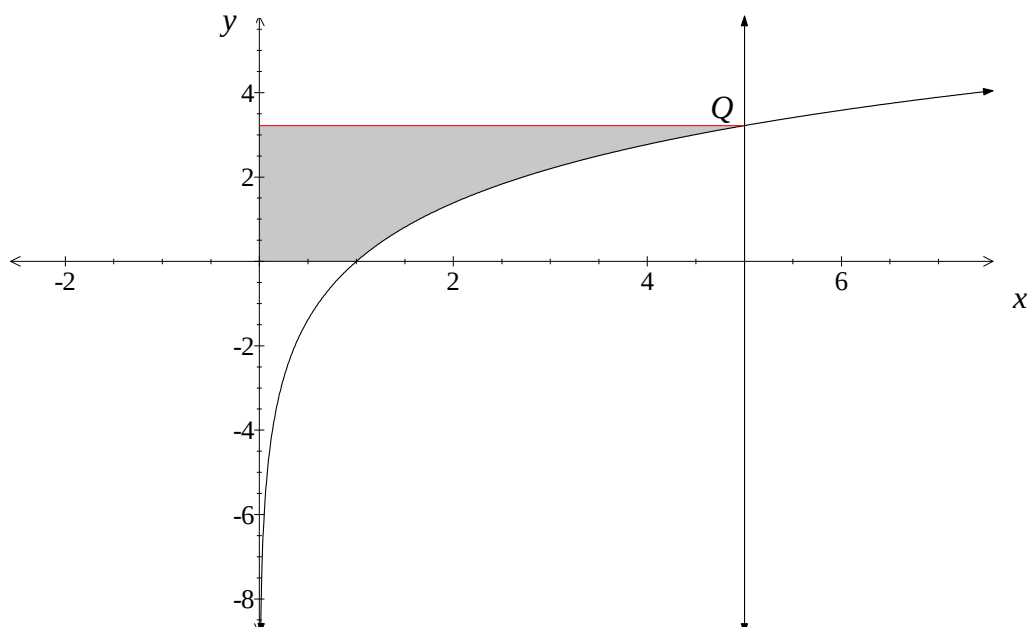
(a) (i) Show that $\frac{5}{\sqrt{5x+3}-\sqrt{5x-2}} = \sqrt{5x+3} + \sqrt{5x-2}$ 2

(ii) Hence find $\int \frac{dx}{\sqrt{5x+3}-\sqrt{5x-2}}$ 2

(b) (i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$. 1

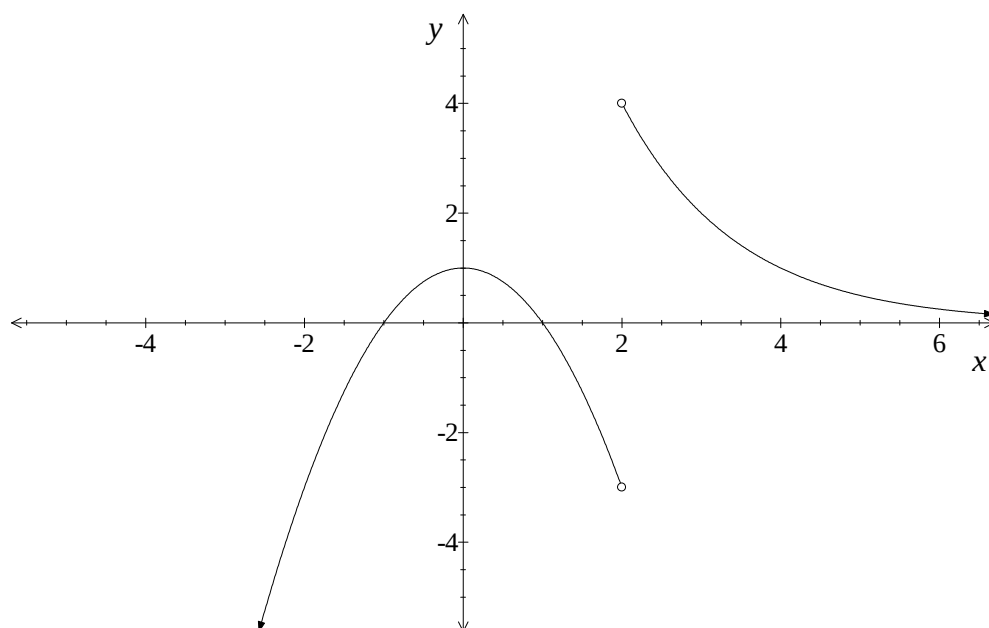
(ii) Hence, or otherwise, find $\int \ln x^2 dx$. 2

- (iii) The graph below shows the curve $y = \ln x^2$ ($x > 0$) which meets the line $x = 5$ at Q . 3
Using your answers above, or otherwise, find the area of the shaded region.

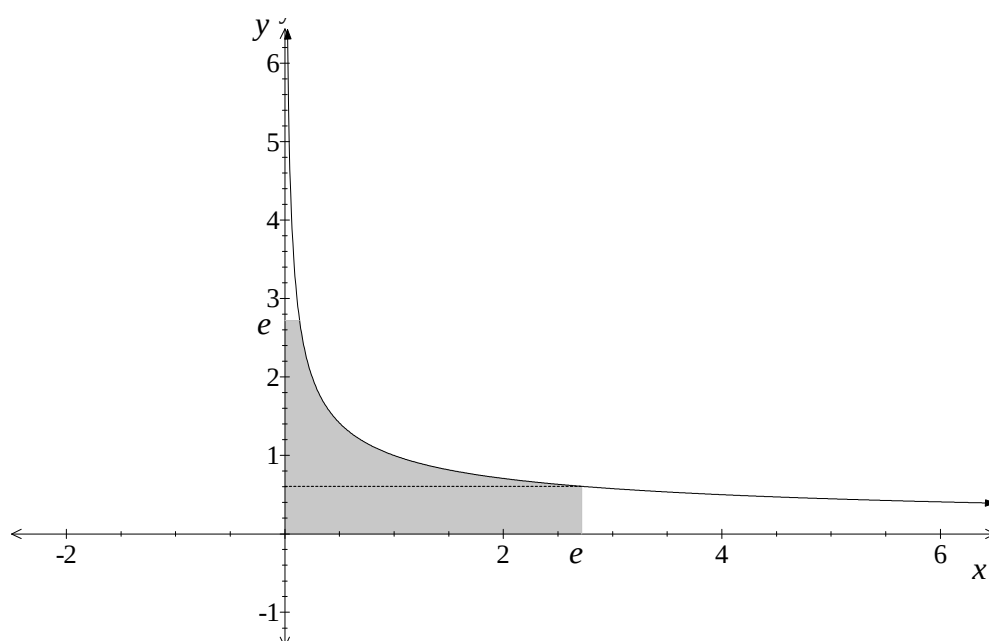


(a) Find $\int \frac{x+1}{x^2} dx$ 2

- (b) The following graph shows the gradient function $y = f'(x)$. 3
 The graph shows that $f'(1) = f'(-1) = 0$.
 Sketch the graph of $y = f(x)$, given that $f'(x)$ is continuous everywhere except at $x = 2$ and that $f(0) = 1$ and $f(-1) = -2$



- (c) The shaded region below is that bounded by $y = \frac{1}{\sqrt{x}}$, the coordinate axes and the lines $x = e$ and $y = e$. 4
 Find the volume when the shaded region is rotated about the y-axis, correct to 2 significant figures.



Question 5 (10 Marks) Start a NEW answer sheet.**Marks**

Consider the function $y = \frac{\ln x}{x}$

- | | | |
|-----|--|----------|
| (a) | What is the domain of this function? | 1 |
| (b) | Show that $\frac{d}{dx}\left(\frac{\ln x}{x}\right) = -\left(\frac{\ln x - 1}{x^2}\right)$ | 1 |
| (c) | Describe the behaviour of the function as x | |
| | (i) approaches zero. | 1 |
| | (ii) increases indefinitely | 1 |
| (d) | Find any stationary points and determine their nature. | 2 |
| (e) | Sketch the curve of this function. | 2 |
| (f) | Hence find the value(s) of k for which $e^{kx} = x$ has no solutions. | 2 |

Question 6	(9 Marks)	Start a NEW answer sheet.	Marks
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- (a) Use mathematical induction to show that the following statement is true 3

$$n^3 + 2n \text{ is a multiple of } 12$$

where n is an even positive integer

- (b) By use of an appropriate diagram and reasons, evaluate the following sum. 2
Do NOT evaluate any primitive functions.

$$\int_0^1 e^x dx + \int_1^e \ln x dx$$

- (c) (i) Show $\frac{1}{u} - \frac{1}{u+1} = \frac{1}{u(u+1)}$ 1

- (ii) Using the substitution $x = \ln u$, find $\int \frac{dx}{1+e^x}$ 3

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$



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ASSESSMENT TASK #3

Mathematics Extension 1

SAMPLE SOLUTIONS

Question 1

- (a) Evaluate $\int_0^1 \frac{dx}{2x+1}$, leaving your answer in the exact form.

$$\begin{aligned}\int_0^1 \frac{dx}{2x+1} &= \frac{1}{2} \int_0^1 \frac{2dx}{2x+1} = \frac{1}{2} [\ln|2x+1|]_0^1 \\ &= \frac{1}{2} (\ln 3 - \ln 1) \\ &= \frac{1}{2} \ln 3\end{aligned}$$

- (b) Using the substitution $u = 4 - x^2$, evaluate $\int \frac{x}{\sqrt{4-x^2}} dx$

$$\begin{aligned}u = 4 - x^2 &\Rightarrow du = -2x dx \\ \int \frac{x}{\sqrt{4-x^2}} dx &= -\frac{1}{2} \int \frac{-2x}{\sqrt{4-x^2}} dx = -\frac{1}{2} \int \frac{du}{\sqrt{u}} = -\frac{1}{2} \int u^{-\frac{1}{2}} du \\ &= -u^{\frac{1}{2}} + c = -\sqrt{4-x^2} + c\end{aligned}$$

- (c) Let $f(x) = \frac{1}{2}(e^x + e^{-x})$ and $F(x) = \frac{1}{2}(e^x - e^{-x})$

Prove that $[f(x) + F(x)]^n = f(nx) + F(nx)$

$$\begin{aligned}f(x) + F(x) &= \frac{1}{2}(e^x + e^{-x}) + \frac{1}{2}(e^x - e^{-x}) \\ &= 2 \times \left(\frac{1}{2}e^x\right) = e^x\end{aligned}$$

$$\text{LHS} = [f(x) + F(x)]^n = (e^x)^n = e^{nx}$$

$$\begin{aligned}\text{RHS} &= f(nx) + F(nx) = \frac{1}{2}(e^{nx} + e^{-nx}) + \frac{1}{2}(e^{nx} - e^{-nx}) \\ &= 2 \times \frac{1}{2}e^{nx} = e^{nx}\end{aligned}$$

- (d) Evaluate $\int_0^1 \frac{e^x}{e^x+1} dx$

$$\int_0^1 \frac{e^x}{e^x+1} dx = [\ln(e^x+1)]_0^1 = \ln(e+1) - \ln(2) = \ln\left(\frac{e+1}{2}\right)$$

Question 2

- (a) Solve $e^x = 5$, leaving your answer correct to 3 decimal places
 $e^x = 5 \Rightarrow x = \ln 5 \approx 1.609437912...$
 $x = 1.609$ [3 dp]

- (b) Find a primitive of $\frac{3x}{1+x^2}$
$$\int \frac{3x}{1+x^2} dx = \frac{3}{2} \int \frac{2x}{1+x^2} dx = \frac{3}{2} \ln(1+x^2)$$

- (c) Find $\frac{d}{dx}(3x \log_e x)$
$$\begin{aligned} \frac{d}{dx}(3x \log_e x) &= 3x \times \frac{1}{x} + 3 \times \ln x \\ &= 3 + 3 \ln x \end{aligned}$$

- (d) Evaluate $\int_0^3 3^x dx$
$$\int_0^3 3^x dx = \left[\frac{3^x}{\ln 3} \right]_0^3 = \frac{1}{\ln 3} (3^3 - 3^0) = \frac{26}{\ln 3}$$

- (e) Using the substitution $u = \log_e x$, evaluate $\int_1^e \frac{(1 + \log_e x)^2}{x} dx$
- | | |
|---|---|
| $x = 1 \Rightarrow u = \ln 1 = 0$
$x = e \Rightarrow u = \ln e = 1$
$u = \ln x \Rightarrow du = \frac{dx}{x}$ | $\begin{aligned} \int_1^e \frac{(1 + \log_e x)^2}{x} dx &= \int_1^e (1 + \log_e x)^2 \frac{dx}{x} \\ &= \int_0^1 (1 + u)^2 du \\ &= \left[\frac{1}{3} (1 + u)^3 \right]_0^1 \\ &= \frac{1}{3} (2^3 - 1^3) = \frac{7}{3} \end{aligned}$ |
|---|---|

Question 3

(a) (i) Show that $\frac{5}{\sqrt{5x+3}-\sqrt{5x-2}} = \sqrt{5x+3} + \sqrt{5x-2}$

$$\begin{aligned}\text{LHS} &= \frac{5}{\sqrt{5x+3}-\sqrt{5x-2}} \\&= \frac{5}{\sqrt{5x+3}-\sqrt{5x-2}} \times \frac{\sqrt{5x+3}+\sqrt{5x-2}}{\sqrt{5x+3}+\sqrt{5x-2}} \\&= \frac{5(\sqrt{5x+3}+\sqrt{5x-2})}{[(5x+3)-(5x-2)]} \\&= \frac{5(\sqrt{5x+3}+\sqrt{5x-2})}{5} \\&= \sqrt{5x+3} + \sqrt{5x-2} \\&= \text{RHS}\end{aligned}$$

(ii) Hence find $\int \frac{dx}{\sqrt{5x+3}-\sqrt{5x-2}}$

$$\begin{aligned}\int \frac{dx}{\sqrt{5x+3}-\sqrt{5x-2}} &= \int \frac{(\sqrt{5x+3}+\sqrt{5x-2})dx}{5} \\&= \frac{1}{5} \int \left[(5x+3)^{\frac{1}{2}} + (5x-2)^{\frac{1}{2}} \right] dx \\&= \frac{1}{5} \left[\frac{1}{5} \times \frac{2}{3} (5x+3)^{\frac{3}{2}} + \frac{1}{5} \times \frac{2}{3} (5x-2)^{\frac{3}{2}} \right] + C \\&= \frac{2}{75} \left[(5x+3)^{\frac{3}{2}} + (5x-2)^{\frac{3}{2}} \right] + C\end{aligned}$$

Question 3 continued

- (b) (i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$.

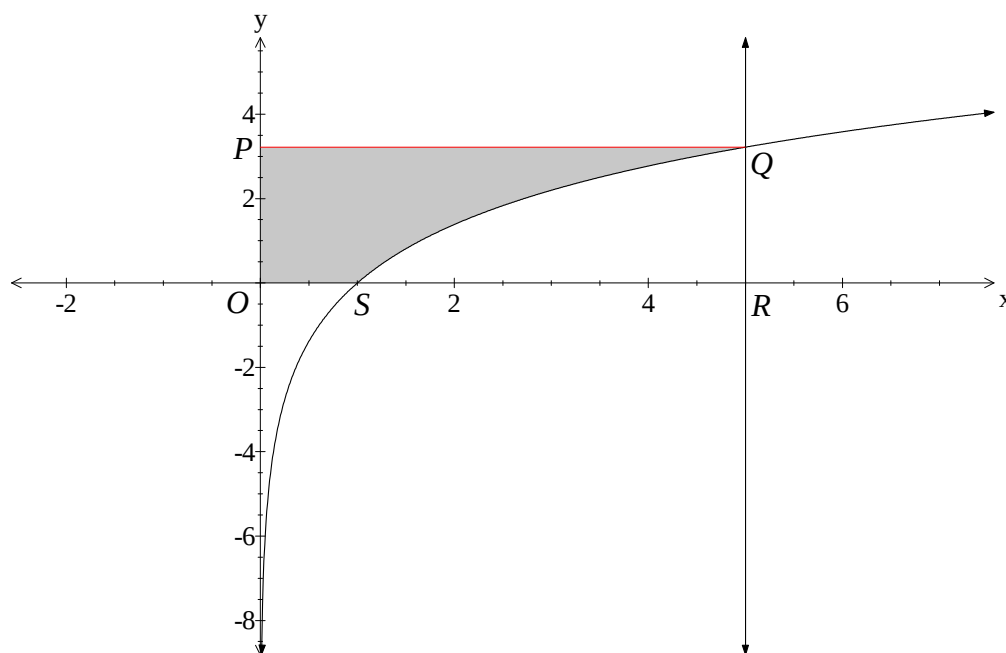
$$\begin{aligned}\frac{d}{dx}(x \ln x - x) &= x \times \frac{1}{x} + 1 \times \ln x - 1 \\ &= 1 + \ln x - 1 \\ &= \ln x\end{aligned}$$

- (ii) Hence, or otherwise, find $\int \ln x^2 dx$.

$$\int \ln x^2 dx = 2 \int \ln x dx = 2(x \ln x - x) + C$$

- (iii) The graph below shows the curve $y = \ln x^2$ ($x > 0$) which meets the line $x = 5$ at Q .

Using your answers above, or otherwise, find the area of the shaded region.



P has coordinates $(0, \ln 25)$

$$\begin{aligned}\text{The required area} &= \text{area rectangle } OPQR - \int_1^5 \ln x^2 dx \\ &= 5 \times \ln 25 - \left[2(x \ln x - x) \right]_1^5 \\ &= 5 \ln 25 - 2[(5 \ln 5 - 5) - (\ln 1 - 1)] \\ &= 5 \ln 25 - 10 \ln 5 + 10 - 2 \\ &= 5 \ln 25 - 5 \ln 25 + 8 \\ &= 8\end{aligned}$$

Question 4

- (a) Find $\int \frac{x+1}{x^2} dx$ 2

$$\int \frac{x+1}{x^2} dx = \int \left(\frac{1}{x} + \frac{1}{x^2} \right) dx = \int \left(\frac{1}{x} + x^{-2} \right) dx$$

$$= \ln|x| - x^{-1} + C$$

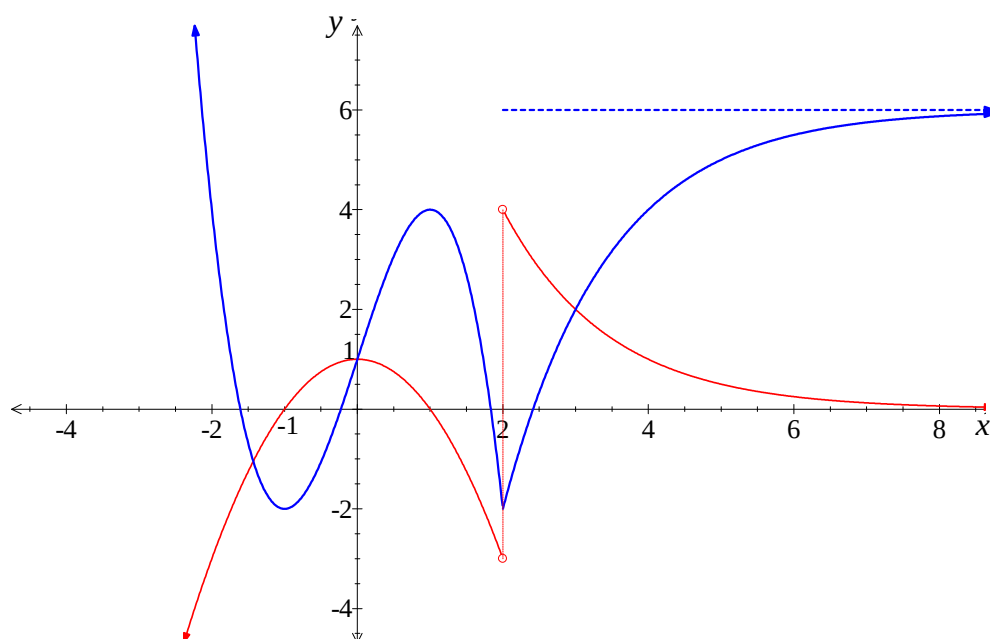
$$= \ln|x| - \frac{1}{x} + C$$

- (b) The following graph shows the gradient function $y = f'(x)$. 3

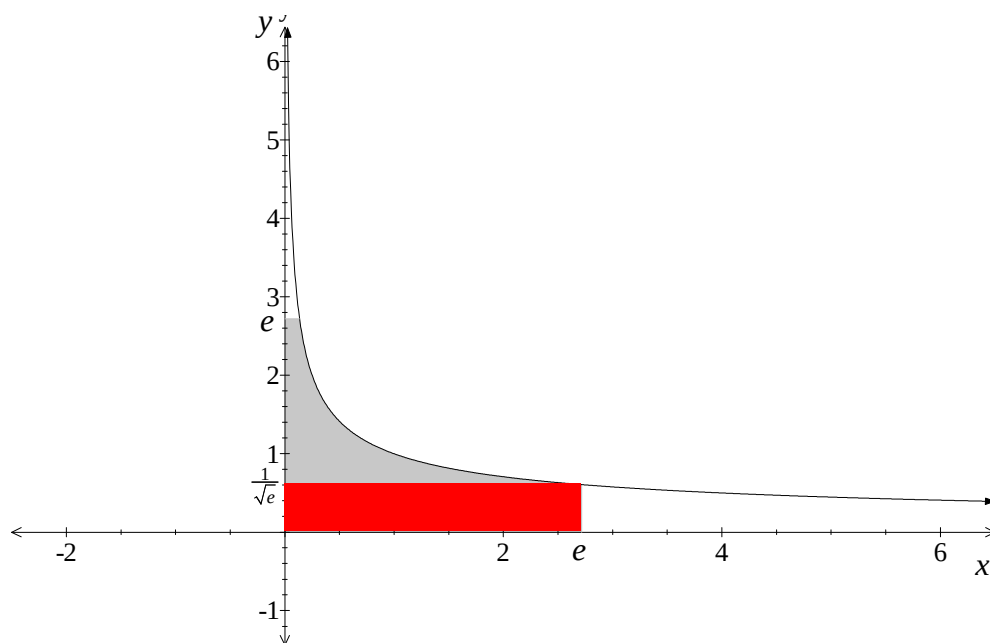
The graph shows that $f'(1) = f'(-1) = 0$.

Sketch the graph of $y = f(x)$, given that $f'(x)$ is continuous everywhere except at $x = 2$ and that $f(0) = 1$ and $f(-1) = -2$

A possible solution:



- (c) The shaded region below is that bounded by $y = \frac{1}{\sqrt{x}}$, the coordinate axes and the lines $x = e$ and $y = e$. 4
Find the volume when the shaded region is rotated about the y -axis, correct to 2 significant figures.



The volume V is the sum of two volumes V_1 and V_2 .

V_1 is the volume formed by rotating the curve $y = \frac{1}{\sqrt{x}}$ from $y = \frac{1}{\sqrt{e}}$ to $y = e$

about the y -axis. $y = \frac{1}{\sqrt{x}} \Rightarrow x = \frac{1}{y^2} \Rightarrow x^2 = \frac{1}{y^4} = y^{-4}$

V_2 is the cylinder formed by rotating the line $x = e$ about the y -axis.

It has radius e and height $\frac{1}{\sqrt{e}}$.

$$\begin{aligned} V_1 &= \pi \int_{\frac{1}{\sqrt{e}}}^e x^2 dy = \pi \int_{\frac{1}{\sqrt{e}}}^e y^{-4} dy \\ &= \pi \left[-\frac{1}{3} y^{-3} \right]_{\frac{1}{\sqrt{e}}}^e = \frac{\pi}{3} \left[-\frac{1}{y^3} \right]_{\frac{1}{\sqrt{e}}}^e \\ &\quad \left[\text{NB } (\sqrt{e})^3 = e\sqrt{e} \right] \\ &= \frac{\pi}{3} \left[-\frac{1}{e^3} + \frac{1}{\frac{1}{e\sqrt{e}}} \right] = \frac{\pi}{3} \left[\frac{\sqrt{e}}{e^2} - \frac{1}{e^3} \right] \\ &= \frac{\pi}{3} \left(e\sqrt{e} - \frac{1}{e^3} \right) \end{aligned}$$

$$\begin{aligned} V_2 &= \pi \times e^2 \times \frac{1}{\sqrt{e}} \\ &= \pi e^{\frac{3}{2}} \end{aligned}$$

$$V = \frac{\pi}{3} \left(e\sqrt{e} - \frac{1}{e^3} \right) + \pi e^{\frac{3}{2}} \approx 19 \text{ u}^3$$

Question 5

Consider the function $y = \frac{\ln x}{x}$

(a) What is the domain of this function? $x > 0$

(b) Show that $\frac{d}{dx}\left(\frac{\ln x}{x}\right) = -\left(\frac{\ln x - 1}{x^2}\right)$

$$\frac{d}{dx}\left(\frac{\ln x}{x}\right) = \frac{x \times \frac{1}{x} - \ln x \times 1}{x^2} = \frac{1 - \ln x}{x^2} = -\left(\frac{\ln x - 1}{x^2}\right)$$

(c) Describe the behaviour of the function as x

(i) approaches zero. $y \rightarrow -\infty$

(ii) increases indefinitely $y \rightarrow 0$

(d) Find any stationary points and determine their nature.

$$y' = 0 \Rightarrow \ln x - 1 = 0 \Rightarrow \ln x = 1$$

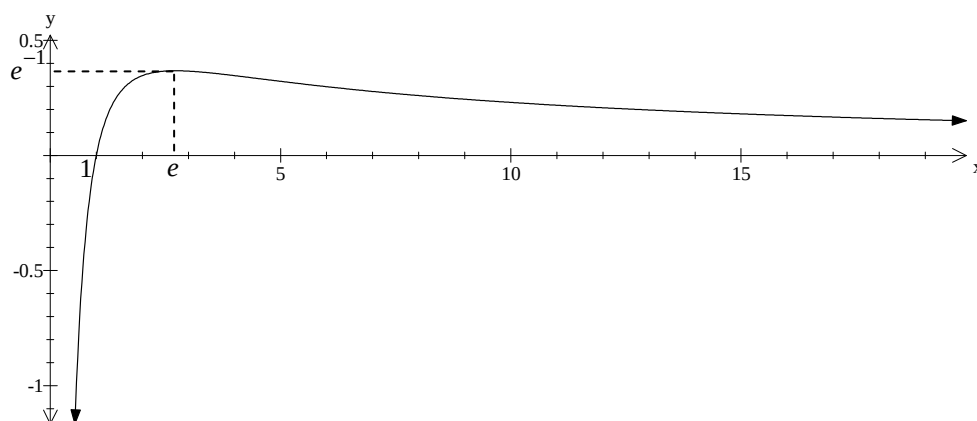
$\therefore x = e \Rightarrow \left(e, \frac{1}{e}\right)$ is the stationary point

x	2	e	3
y'	0.3	0	-0.1

Only need to check $(1 - \ln x)$ as $x^2 > 0$.

So (e, e^{-1}) is a maximum turning point.

(e) Sketch the curve of this function.



(f) Hence find the value(s) of k for which $e^{kx} = x$ has no solutions.

$$e^{kx} = x \Rightarrow kx = \ln x$$

$$\therefore k = \frac{\ln x}{x}$$

So the solutions to $e^{kx} = x$ are found by intersecting the line $y = k$ with

$$y = \frac{\ln x}{x}.$$

So there will be no solutions when $k > \frac{1}{e}$.

Question 6

- (a) Use mathematical induction to show that the following statement is true

$$n^3 + 2n \text{ is a multiple of 12}$$

where n is an even positive integer

Test $n = 2$

$$2^3 + 2 \times 2 = 12$$

Clearly $n = 2$ is true.

Assume true for $n = 2k$ i.e. $(2k)^3 + 2(2k) = 12N$, $N \in \mathbb{Z}$

$$\therefore 8k^3 + 4k = 12N.$$

NTP true for $n = 2k + 2$ i.e. $(2k + 2)^3 + 2(2k + 2) = 12M$, $M \in \mathbb{Z}$

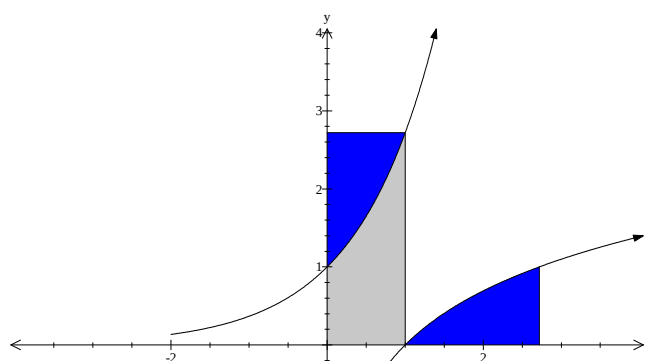
$$\begin{aligned} (2k + 2)^3 + 2(2k + 2) &= 8(k + 1)^3 + 4(k + 1) \\ &= 8(k^3 + 3k^2 + 3k + 1) + 4k + 4 \\ &= (8k^3 + 4k) + 24k^2 + 24k + 12 \\ &= 12(N + 2k^2 + 2k + 1) \\ &= 12M \quad \left[\because N + 2k^2 + 2k + 1 \in \mathbb{Z} \right] \end{aligned}$$

So the statement is true for $n = 2k + 2$ provided it is true for $n = 2k$.

So by the principle of mathematical induction it is true for all positive even integers.

- (b) By use of an appropriate diagram and reasons, evaluate the following sum.
Do NOT evaluate any primitive functions.

$$\int_0^1 e^x dx + \int_1^e \ln x dx$$



By symmetry the integral $\int_1^e \ln x dx$ produces the same area as that of e^x next to the y -axis for $1 \leq y \leq e$.

So $\int_0^1 e^x dx + \int_1^e \ln x dx$ is the area of the rectangle with dimensions $1 \times e$

$$\therefore \int_0^1 e^x dx + \int_1^e \ln x dx = e$$

Question 6 continued

(c) (i) Show $\frac{1}{u} - \frac{1}{u+1} = \frac{1}{u(u+1)}$

$$\frac{1}{u} - \frac{1}{u+1} = \frac{u+1-u}{u(u+1)} = \frac{1}{u(u+1)}$$

(ii) Using the substitution $x = \ln u$, find $\int \frac{dx}{1+e^x}$

$$\begin{array}{l|l} \begin{array}{l} x = \ln u \Rightarrow dx = \frac{du}{u} \\ x = \ln u \Rightarrow u = e^x \end{array} & \begin{array}{l} \int \frac{dx}{1+e^x} = \int \frac{1}{1+e^x} \times dx = \int \left(\frac{1}{1+u} \right) \frac{du}{u} \\ = \int \left[\frac{du}{u(u+1)} \right] = \int \left(\frac{1}{u} - \frac{1}{u+1} \right) du \\ = \ln u - \ln(u+1) + C \\ = \ln \left(\frac{e^x}{1+e^x} \right) + C \quad \quad \quad \left[= x - \ln(1+e^x) + C \right] \end{array} \end{array}$$

End of Solutions